

Wheat Based Ai Disease Monitoring and Detection

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Abstract— Wheat is one of the most important food grains in the world. Rust infections are one of the most common diseases affected on wheat leaves. This paper discusses about detecting anomalies on wheat leaves (yellow, healthy and brown) using isolation forest algorithm, implementation of teaching a MobileNetV2 architecture-based deep learning model to categorize photos of wheat crop into several groups, look for abnormalities in crop photographs that differ from typical disease makers such as pustules, lesions as these could be signs of infection, plotting of bounding boxes around rust infections. YOLO algorithm is also implemented here. It uses yolov3 a pretrained YOLO model to perform object detection in an input image.

Keywords— *isolation forest, anomaly detection, yolo, yolov3, wheat rust infection, bounding boxes, supervised, unsupervised learning.*

I. INTRODUCTION

Wheat is one of the most significant staple grains in the world, Wheat is the main source of nutrients and calories produced from plants in diets for humans. Greater crop output is required to achieve the “hidden hunger” objectives due to changes in grain quality criteria and increased use of wheat based goods in many Asian countries. Challenges to the cultivation of wheat both nationally and internationally include diminishing amount of viable agricultural terrain, warming temperatures and a range of erratic abiotic and biotic pressure. High performing cultivars has led to a decrease in wheat’s genetic diversity, which has created a perfect storm for pathogen development to the point where diseases are a threat to world’s wheat supply[1]. Nuemorous rust disease kinds typically harm this crop. One type of fungal infection that affect wheat crop is rust. There are three types of rust : yellow (Stripe rust), black(stem rust) and brown (leaf rust). These kinds of illness have the potential to cause enormous loses in wheat productivity[2].

Wheat commerce surpass all other combined trade and it is the second most produced cereal grain after maize. 760 million tons of wheat were produced worldwide in 2020. Together Russia, India and China are the world’s top three wheat producers, contributing over 41% of wheat production. In terms of total wheat production, the US ranks fourth in the world. Wheat is the main vegetable source in human meals, with a protein level of roughly 13%. It is a significant source of carbs as well. Whole grains such as wheat are a great source of minerals and dietary fibre. Gluten, a protein found in wheat can harm the tiny intestines of people with celiac disease- an immune system condition that is inherited. Additionally,

gluten can cause dermatitis, herpetiformis, gluten ataxia and non-celiac gluten sensitivity[3].

Plant disease is a transition from a healthy state to a sick state that may alter or interfere with a plants ability to perform essential functions. Every species of plant, domesticated or wild is affected by the illness. Plant diseases are caused by bacteria, viruses and fungi which are pathogenic agents. Living things like rodents, fleas are considered as pests. These can be caused by two factors: abiotic and biotic[2]. Biotic factors are nothing but living organisms such as bacteria, animals and plants. Abiotic factors are non-living elements like soil, atmosphere and water [4]. Biotic can infect any kind of leaf, root stem and may propagate from crop to crop. All of the living components of an ecosystem are known as biotic services. Abiotic factors encompass all elements of an environment that are not living. These are the chemical agents- gas, nutrients and soil as well as the physical parameters such as different temperatures needed for different crops, pH values and dew ratios[2].

Rust infections are prevalent in wheat growing regions and are considered the most economically important bacterial infections affecting wheat and other cereal crops globally. Under ideal meteorological circumstances, rust illness can grow quickly and have the capacity to disperse windborne spores over great distances. Stripe, stem and leaf rust are the three types of rust infections that affect wheat. In the middle plain regions and other American regions that cultivate wheat, leaf rust is among the most prevalent of 3 illness. The leaf rust is an annual occurrence in certain states. Since many forms of stem rust have become immune to disease, it is usually less common than other rusts with the current outbreaks in the state along the great plains, stripe rust is a disease that is becoming more and more significant.

- i. **Stripe rust (Yellow rust):** *Puccinia striiformis* f.sp. *tritici* is the fungus that causes yellow rust, which is also called stripe rust. The disease mostly affects leaves though it can also affect awns and glumes. This age of life cycle in “hot spots” of infection that appear in crops in the early spring and later winter. Usually orange or yellow blister-like pustules grouped in stripes are the result of stripe rust. Massive volumes of easily detached and wind-dispersed spores are produced by lesions. People strolling through badly infected fields may find orange dust on their clothing from the spores[5].



Figure 1 .[6]

- ii. **Leaf rust(Brown rust):** *Puccinia triticina* is the fungus that causes leaf rust, commonly referred as brown rust. Whenever barley, wheat and other grains are planted this rust disease can appear. Leaf rust solely affects foliage. Dusty, an orange-red to brownish-red fruit bodies that emerge on the surface of leaves are the telltale signs. Countless spores are produced by these lesions, and can completely cover the upper leaf surface. Water splashing and wind disperses the spores of leaf rust [5].



Figure 2 [6].

- iii. **Stem rust(Black rust):** *Puccinia graminis* f.sp.tritici is the fungus which causes stem rust. Although they can also occasionally form on glumes, seeds and awns elevated spots on branches and leaves with sheaths are indicative of stem rust. Stem rust spore are more extended than stripe and leaf rust spots, giving them an elliptical appearance. The dots, which range in colour from orange to dark red, appear on the top and lower leaf surfaces. Stem rust patches have jagged edges. Many spores are released by immature pustules. Black rust gets its common name from the transformation of spores later over the season of growth, which turns them dark in colour[5].

Five distinct spore stages and two host-wheat, the primary host and an alternate host are part of intricate life cycles of wheat rust[7]. Nonetheless, because the other host and the three out of five spores phases have minimal bearing on the development of in-season rust. Ohio's in-season rusting cycle is a pretty simple one . The spore type that spreads and infect the wheat crop is uredospore's stage. It reproduces and attack leaves in six to eight hours soon after touching the plant surface when temperatures and moist levels are ideal. If the climate is conducive, uredospore's can generate a new crop in seven to fourteen days. There is a higher chance of severe

outbreaks and loss of crop during the season along with additional spores and the infection cycles that occur for sooner rust develops. The best condition for the formation of leaf rust are light rain, high levels of humidity and temperatures between 77 and 86F. Up until the crop reaches maturity, wind-blown uredospore's travelling crop to crop and region to region propagate all three illness. Depending on rust, black submerged pustules may appear in surface of leaves, spikes , leaf sheaths and stems as the plant ages. The winter spores or teliospores are found in these pustules or telia. Wheat is not infected with teliospores when plants contract the disease relatively late in the growing season nearing the maturity, telia may not emerge. Uredospore's travel southward in autumn, infecting wheat and surviving the winter as mycelium or uredospore's on volunteer wheat plants.

There are certain management ways in which this can be reduced:

- i. Switching to varieties in the event of rust epidemic
- ii. Autumn infections are more likely when plants are planted early. Serious infections causes stress and increase the risk of winter damage in seedlings.
- iii. A soil test based, balanced fertilisation programme should learn the likelihood of significant yield loss from rust. Elevated nitrogen rates in the absence of adequate P and K can worsen rust, which lowers grain production.
- iv. There are fungicides available to control wheat leaf rust. When the kind of plant is vulnerable, the illness began early. Flag leaves has a threat of contracting, using a specific fungicide is frequently advised[7].

Look for abnormalities in crop image that deviate from typical illness makers such as lesions, pustule or discolouration. Determine and categorize cases of wheat rust infection with precision and accuracy. Defining forms of wheat rust infection by placing bounding boxes around areas of interest. Data selection preprocessing, model selection, training, evaluation and deployment are the solution approach that is used to achieve all the above goals. Early detection of wheat rust result in increasing crop yield and decreased economic loss. Precise detection and classification of wheat rust disease enables farmers to apply focused treatments only when required, maximising resource efficiency and decreasing the demand for extensive pesticide use. Ability to automatically identify anomalies in crop pictures allows for ongoing field monitoring. The dataset used for achieving the goal is Wheat Rust Classification Dataset (WRCD). The dataset can be achieved by using following link : <https://www.kaggle.com/sinadunk23/behzad-safari-jalal>. The dataset consist of three classes : Healthy, Brown rust and Yellow rust. Brown rust consist of 1128 images indicating images of leaves affected by brown rust. Yellow rust consist of 1156 images indicating leaves affected by yellow infection and last class healthy consist of 1395 pictures indicating leaves that are healthy [8].

II. BACKGROUND RESEARCH

What is anomaly detection?. The process of discovering observations, occurrences or data points that

differ from the norm, standard or expected and are the inconsistent with the reminder of the dataset is known as Outlier detection or Anomaly detection. ML(machine learning) and AI(artificial intelligence) are used to automatically detect unexpected deviations in a datasets typical behaviour. There are numerous applications for anomaly detection in different sectors. It is used for instance, in cybersecurity to identify anomalous network activity, in manufacturing to identify flaws or equipment failures, in finance to detect fraud and in healthcare for discovering aberrant medical conditions. Anomaly detection is important because of various reasons: **enhanced data quality** is necessary for accurate and trustworthy data analysis, and it can be greatly enhanced by recognising and managing data abnormalities. **Improved decision making** is another important reason. **Optimal execution** of ML algorithms can be achieved by avoiding data anomalies which lead to model fitting the noise instead of identifying the true pattern in the data. There are various methods for anomaly detection they are: **Visualisation** – data scientists may spot possible abbreviations and trends in information by using visualisation, which makes it an effective tool for identifying anomalies in data. Analyst can visually check the dataset for any odd data points or trends by displaying the information using graphs and charts. Statistical tests are another method for anomaly detection such as Grubbs test, Kolmogorov-Smirnov test and so on. ML algorithms such as SVM (support vector machine), Naïve Bayesian, KNN, Autoencoders, decision trees, K-means clustering are methods used [9].

Alongside by data technology, high performance, computing, ML has arisen to open up new avenues for interpreting, quantifying and comprehending data-intensive operations in agricultural environment, yield production is one of the most prominent applications of ML in agriculture, that is another key use case with samples for tomatoes, corn, potatoes, beans maize and so on. Because ML can identify intricate correlation between a wide range of elements and eliminate irrelevant ones, it is an excellent tool for this purpose. To decide whether to apply a crop protectant or any other control measures, numerous decision support systems (DSS) regarding protecting crops are utilised. This method can be conceptualised as an ML classification issue, in which crops are classified as healthy or diseased based on input data and model is then developed to forecast the likelihood of illness. In a traditional classification task, everything is expressed as a vector of features and information representing all classes of interest are accessible [10].

Using deep anomaly, deep learning models were used in agriculture. Deep anomaly detects object impediments and abnormalities in a farming region by combining deep learning with background subtraction. It makes use of uniform visual qualities and variety of barriers found in agricultural fields. Finding distant and highly hidden objects, as well as categories of unknown objects is the main objective [11]. Deep learning has been extensively researched and used for fungal spore detection and diagnosis due to its benefits of being inexpensive and great efficiency which have been made possible by advancements in computer hardware performance. Additionally, a variety of neural networks have been used to identify fungal illness from photographs of minute spores[12]. The YOLO(you only look once) group of algorithms, which combines high accuracy and speed, are being applied in many scene detection jobs as object detection software has advanced. The yolo system predicts every object

and computes every characteristics in the image simultaneously. The 5th iteration of YOLO was created using programming language of python [13]

III. METHODOLOGY

The primary goal is to use ML to improve crop disease control strategies, which will ultimately increase wheat production and guarantee food safety. Isolation forest algorithm is used to achieve the primary objectives of this paper. The main objectives are to find unusual patterns or deviations from the normal conditions of the crop by anomaly detection algorithm. To examine crop photos for anomalies that deviate from conventional disease indicators like pustules, lesions which are signs of an infection. To identify and classify wheat rust infection cases precisely and accurately using crop photos and to draw bounding boxes representing varieties of wheat rust illness [15].

Isolation forest is an anomaly detection algorithm. This approach in unsupervised learning isolates outliers from the data to identify anomalies. The algorithm known as decision tree is the foundation of isolation forest. Its primary uses are regression and classification. Isolation forest yields better results when used with RGB and HSV color schemes. The isolation forest is employed to distinguish a small number of distinct datapoints among other comparable datapoints in the datasets. The primary benefit of using an isolation forest cannot be about calculating distances, but rather about working nonstop until anomalies within the data are found [14]. By choosing a feature followed by choosing a split value based on features minimum and maximum scores at random, it segregates the outliers. The unusual data points will be separated from the other parts of the information by the shorter pathways that result in random feature division. Creating a profile of what is normal is typically the first stage in anomaly detection process. Something which couldn't be deemed normal is then reported as abnormal. Nevertheless, the isolation forest does not operate according to this theory, it does not compute point-based distances or define normal behavior first. The foundation of isolation forest method is the idea that observations that are scarce and distinct from one another constitute anomalies, which ought to make them simpler to spot. To identify anomalies, isolation forest employs a collection of isolation tree based on provided data points [15].

Creating a group of random trees using a certain collection of data points having the smallest paths are called anomalies. It then assumes that any dataset outliers are easier to extract than normal points. Every points anomaly scores are determined by using the following equation:

$$2^{-E(h(x))/c(n)} \quad (1)$$

Where $h(x)$ denotes the count of edge to a point x of a tree and $c(n)$ denotes constant of normalization for a size n dataset. An isolation value can be applied in unsupervised or supervised environment. Score analyzation can be done in the following manner. A score nearer 1 will be assigned to anomalies. Significantly lower than 0.5 indicate that there don't appear to be any glaring anomalies in the sample as a whole [16].

Operations involving images on the most basic degree of abstraction. Input and result are intensity images and are commonly referred to as pre-processing. Preprocessing aims to improve the image data by suppressing unwanted distortions and enhancing certain image features that are crucial for subsequent processing. However, because comparable approaches are used, geometric image transformations such as scaling, translation and rotation are included in the preprocessing methods category [17]. An image that solely consist of grayscale also known as gray level colors is called gray scaled image. These photos are distinguished from the ordinary color images by the fact that each pixel requires fewer details to be delivered.

Every picture in a grayscale image contains a value from 0 to 255, with 255 denoting white and 0 denoting black. Grayscale values ranges from 0 to 255, with values that are near 25 being lighter and those closer to 0 being darker [18]. An RGB image can be converted into grayscale by following methods:

- i. **Lightness method:** Finding the mean value for the elements with the greatest and smallest values is a pretty straightforward method. The equation is as follows:

$$\text{grayscale} = \frac{\min(R,G,B) + \max(R,G,B)}{2} \quad (2)$$

This method's use of only one RGB component creates a very major flaw. This is a serious issue since our perception of brightness is based on all three of primary colors.

- ii. **Average method:** It takes the mean values of every three components of RGB. It is denoted by the equation:

$$\text{grayscale} = \frac{R+G+B}{3} \quad (3)$$

The average technique gives each component the same weight, it is problematic.

- iii. **Luminous method:** It is the best technique. It is expected that green will contribute more to result, and blue will contribute less. Following several tests and a more through investigation, scientists came to the following conclusions:

$$\text{grayscale} = 0.3 * R + 0.59 * G + 0.11 * B \quad (4)$$

Among all the three methods luminous is considered as the best [19].

Discussing about the necessity of gray scaling, gray scaling images typically are considered as single-dimensional, but RGB images have three color streams thus three dimensions and this result in dimensional reduction. Gray scaled image is used for minimizing the complexity of the model. For example, neural module training on 10 x 10 x 3 RGB images. There will be 300 nodes for input in input layer. However,

with grayscale photos, a similar neural network model will require just 100 input nodes. A lot of algorithms are tailored to only function with grayscale images. For example, the OpenCV libraries pre-implemented canny edge detection algorithm only operates with grayscale images [20].

Feature extraction here is done by HOG. HOG is nothing but histogram of oriented gradient. A feature descriptor called HOG counts the instances of orientation gradients in specific areas of pictures. The image is divided into small areas called cells, and localized contrast in overlapped blocks is normalized after histogram of gradient pattern is computed inside every cell. The gradient of intensities of pixel in every direction is calculated both vertically and horizontally. Convolution in the X and Y directions via kernel of [-1, 0, +1] is comparable. Eventually, each pixel will have a Gx and Gy scores. The gradient magnitude and angle are computed using the values derived from the Gx and Gy. The angles indicate the direction of the edge, while the gradient magnitude indicate how strong it is.

$$g = \sqrt{g_x^2 + g_y^2}$$

$$\theta = \arctan \frac{g_y}{g_x} \quad (5)$$

For each cell, HOG can be calculated by separating the images into cells, which are small, connected areas for example, eight by eight pixels. Creates a gradient histogram for each cell. Splitting of orientations over nine bins, every one with a 20-degree angle is represented by the histogram bins. Calculate the total gradient magnitude of each pixel in corresponding bin that has a gradient angle. There will be nine gradient magnitude values in each cell's histogram. The histograms are normalized over a larger, overlapping region known as blocks to offset irregularities brought on by fluctuating noise and lightning. Every block can include more than one cell for example 2 x 3 cells which constitute of 16 x 16 pixels. Combine all of cells HOG into a single array, then use L2 norm to normalize it. This computation uses overlapping blocks with a single cells stride. Assemble all the images blocks HOG features, then join them to create a composite feature vector that represents the images HOG features collectively. The total amount of characteristics with HOG extraction of features for an image in M x M dimension considering M as a gradient angle bins with twenty degrees, a block dimension of 2 by 2 cells, a cell dimension of 8 by 8 and a stride of 1 cell will be:

$$\left(\frac{M}{8} - 1\right)^2 \times 36. \quad (6) \quad [21].$$

Flattening is another technique used here the purpose of flattening which turns arrays with multiple dimensions into a single-dimensional array and is typically employed in deep learning to feed the classification model with the information from the one-dimensional arrays. The primary rationale for flattening an image array prior processing or feeding the data to our model is because 1-D arrays require less memory than multi-dimensional arrays [22].

Autoencoders are a unique kind of deep learning framework and is utilized to learn data

representations only through descriptive feature-based learning. The goal of learning the form of representation, essentially constitutes a change of initial information to most closely recover the raw information. Denoising, generative applications, and dimensionality reduction can all be accomplished using this representation technique. It has two parts: encoder and decoder. Encoder transforms the data provided into representations, which is sometimes known as bottleneck or code. Decoder that converts the code into an input reconstruction. By employing the error in reconstruction as an indicator, an autoencoder could be used to stimulate the typical behavior of data and identify outliers. Specifically, training it to fully reconstruct events in a dataset is one method to achieve this. In wheat rust classification, autoencoders are used as an unsupervised learning for anomaly detection [23].

$F(x)=h$, where x provides the data that is input and h provides the latent variable that are present in the data bottleneck, is a general mathematical equation for defining an autoencoder. The encoder portion of network is indicated by this formula. The information bottlenecks latent variables are taken by the decoder, who then maps them into an output that may be expressed as $g(h)=x$. usually the decoder is the encoders mirror image. The autoencoders can be expressed as:

$$L(x, g(f(x))) \quad (7)$$

Where g and f are both functions that are nonlinear [24].

IV. EVALUATION

To detect unusual patterns or deviations from normal conditions of the crop, isolation forest algorithm is used. Color images are converted to grayscale using grayscale method. These algorithms efficiency and accuracy can be increased with the aid of gray scaling [25]. For anomaly identification, isolation forest algorithm is employed. Based on their characteristics, it builds isolation trees to separate anomalies from regular observation [26]. To capture the structure and shape of local objects, characteristics called HOG are retrieved from the gray scaled image. Anomaly detection and other computer vision tasks make use of these attributes [21]. The importance of contamination parameter indicates the percentage of abnormalities inside dataset [27]. By comparing projected anomalies with the ground truth labels and calculating the percentage of accurate prediction accuracy is determined.

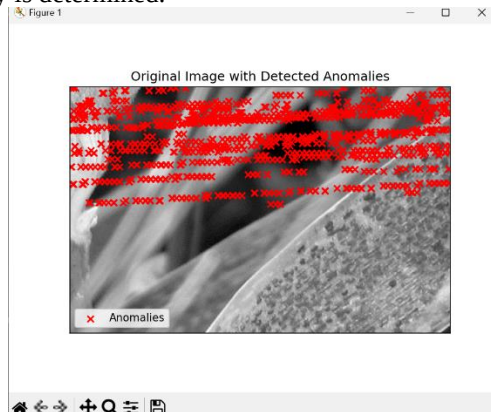


Figure 3

The above figure represents anomalies (red marks) detected by isolation forest algorithm on brown rust with an accuracy of 99%, which means that 99% of anomalies were correctly identified by isolation algorithm. Anomalies were detected in dataset. The array [448, 449, ... ,120730] represents indices of anomalies in dataset which represent positions of anomalous instances within the dataset. It is depicted in figure4.

```
Detected anomalies: [ 448 449 451 ... 120722 120729 120730]
Accuracy: 0.9908112642459581

Process finished with exit code 0
```

Figure 4

Similarly the effectiveness of the isolation forest algorithm in identifying abnormalities across various classes of images(healthy, brown and yellow rust) in dataset were performed and produced an accuracy of the following:

```
C:\Users\devan\Masters_project\.venv\Scripts\python.exe C:\Users\devan\Masters_project\isol1.py
Accuracy for Healthy: 0.0
Accuracy for Brown Rust: 0.99
Accuracy for Yellow Rust: 0.9902884615384615
```

Figure 5

While isolation forest method does not find any anomalies in healthy class, it did a great job in identifying anomalies connected to rust, brown and yellow rust. Rust classes have high accuracy scores which suggest that anomaly detection is successful for these classes. The plot of this is depicted in figure6.

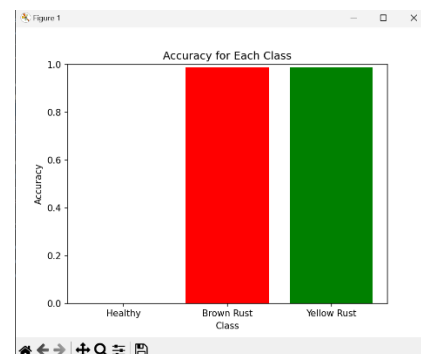


Figure 6

To detect anomalies in crop photographs that differ from typical disease indications, such as sores, discolouration which indicate an infection. The 'load_dataset' loads images from given path resize the images to target size, label it according to whether or not they are anomalies. Preprocessing is done by flattening an isolation forest model and is trained on flattened images to detect anomalies. It predict anomalies, recognise anomalies using trained model prediction. The evaluation metrics of recall, F1 score and precision is calculated. The algorithm detects 30 anomalies that deviate from normal conditions of crop. It is depicted in figure7.

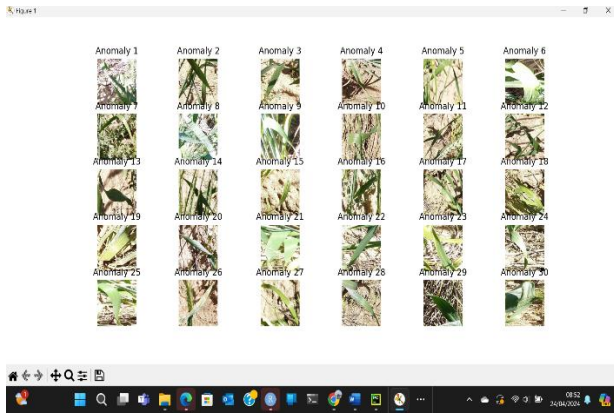


Figure 7

Recall defines a classification model capacity to recognise every data item in a permanent class. The equation of recall is

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (8)$$

Precision is the capacity of a system of classification to yield only the class specific data points. The equation is:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (9)$$

F1 score is nothing but the individual measure that use harmonic mean to integrate precision and recall. It is denoted by:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

The ground truth labels (y_{true}) and the list of anomaly indices found by the isolation forest model (anomaly indices) are passed to evaluate_anomaly_detection function. Based on these inputs, it computes the true positives, false positives and false negatives. Then using the previously described formulas, it computes the precision, recall, and f1 score. Ultimately the function returns these metrics, which can be utilised to access how well the anomaly detection method is working. The displayed Precision, Recall and F1 score is represented in Figure 8.

```

C:\Users\devan\Waters_project1\venv\Scripts\python.exe C:\Users\devan\Waters_project1\isolation_forest2_notif.py
Precision: 0.5
Recall: 0.125
F1-score: 0.2
Number of anomalies detected: 30

```

Figure 8

Precision is 0.5 which means that 50 % of them are TP. Recall is 0.125 which means model correctly identified 12.5% of actual anomalies. F1 score is 0.2 which indicate a moderate performance [28].

Make exact reliable identification and classification of wheat rust infection instances with crop images in done. For preprocessing and data augmentation,

‘imageDataGenerator’ object called train-dataset and validation_dataset are utilised, which alters the image in number of ways such as zooming, rotation. Preprocessing and real-time data augmentation of picture data are done with the help of imageDataGenerator class. In order to give the model a huge number of different photos without storing them all in the memory, it generates groups of enhanced image instantly throughout training [29]. MobileNetV2 is used as a base model. It is a CNN(convolutional neural network) with 53 layers. The ImageNet database contains an initial version of the network that is trained on a million photos[30]. To suit the particular classification problem, two further layers named GlobalAveragePooling2D and Dense are added to the base model. The final class predictions are generated by Dense layer using SoftMax activation after the Global AveragePooling2D layer lowers the spatial dimensions for the data mapping to a vector of fixed size. By assigning False to its trainable attribute, the base model layers are rendered immobile. Because of this, only the recently added layers may be trained during training because their weights cannot be changed. The categorical cross entropy function of loss and Adam optimiser are used to compile the model[31]. Training the model is done by ‘fit’ method. The output is depicted in figure 9.

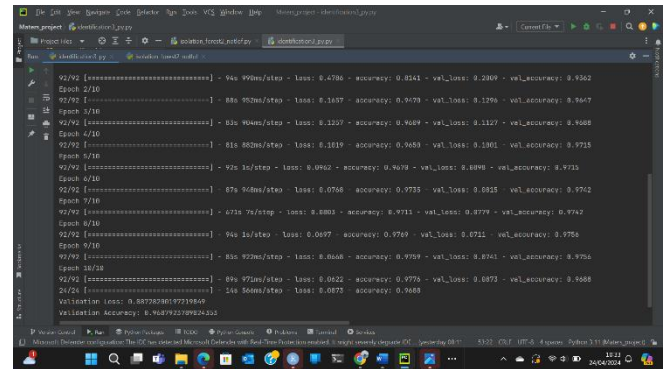


Figure 9

The model is trained for 10 epochs with each iteration over entire training dataset. Using validation and training datasets models performance is evaluated. It shows the accuracy and loss metrics for the validation plus training sets at each epoch. Low value of loss represents that model makes meaningful results. Accuracy increases over each epoch indicating model is becoming good at classifying the images accurately. It shows effective learning as there are improvements in both validation and training metrics. The model is tested with the validation set to determine how well it performs on unseen data after training. It shows a validation loss on the validation dataset and validation accuracy is 96.88% which means it accurately classifies about 96.88% of images in validation set. Overall the system produced great results with low validation loss and a high validation accuracy.

Placing bounding boxes around regions of interest to symbolise various forms of wheat rust infections are also done. Preprocessing steps such as thresholding, Gray scaling, erosion and dilation are also done. Bounding boxes are superimposed on original image to visually represent wheat rust infection. A bounding box will be used to outline each region that is detected. It is represented in figure 10.

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